

# NOTIUNI FUNDAMENTALE .NET/C#

**Tehnici avansate de programare**

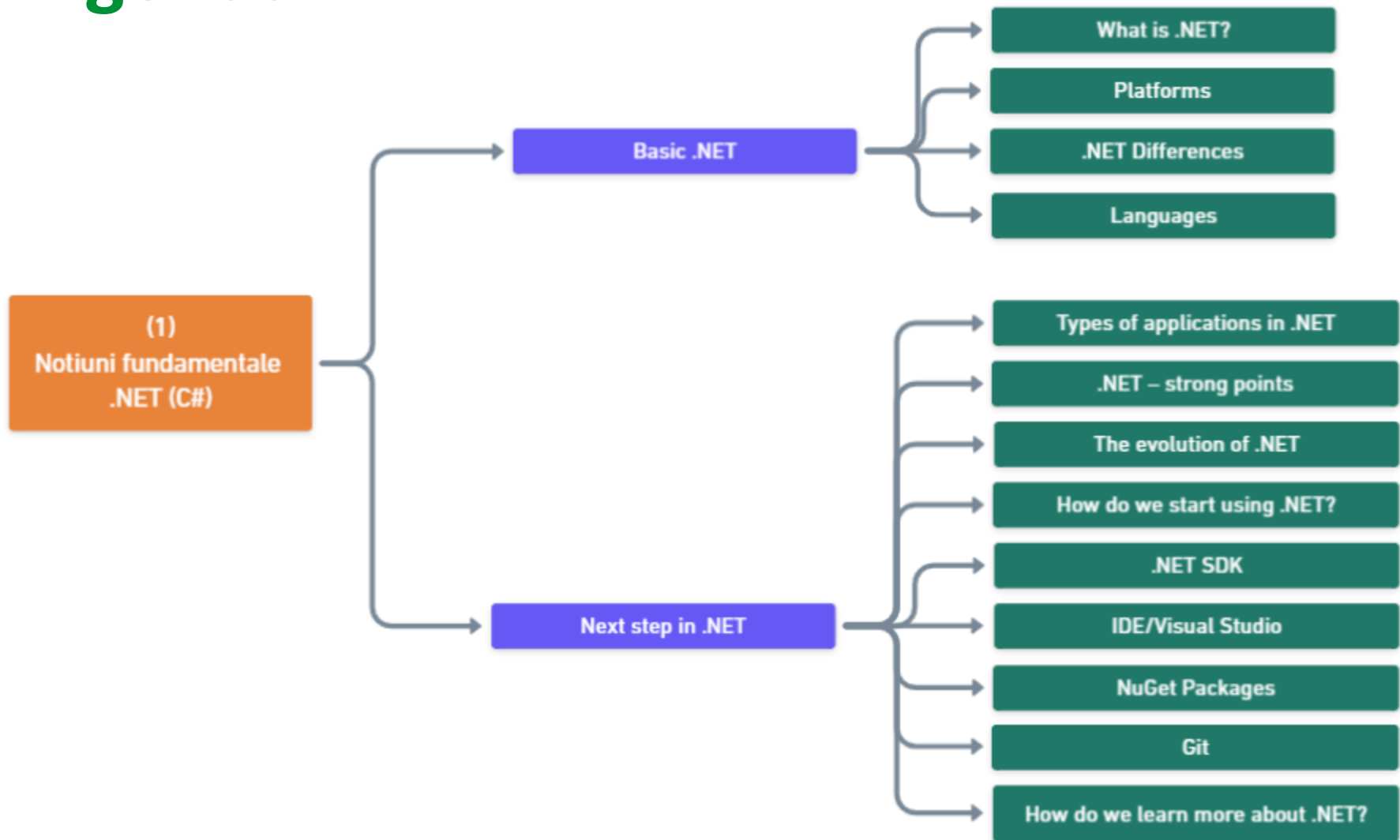
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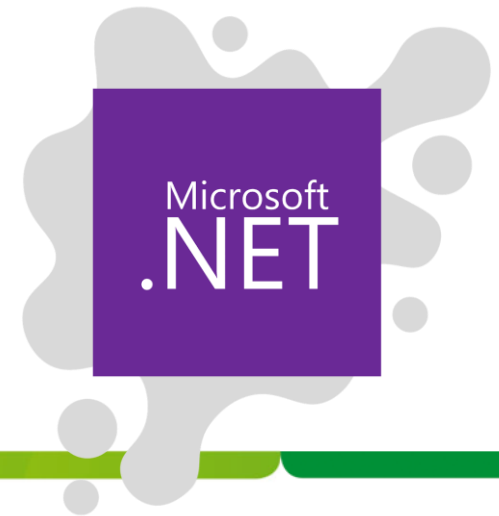


# Agenda



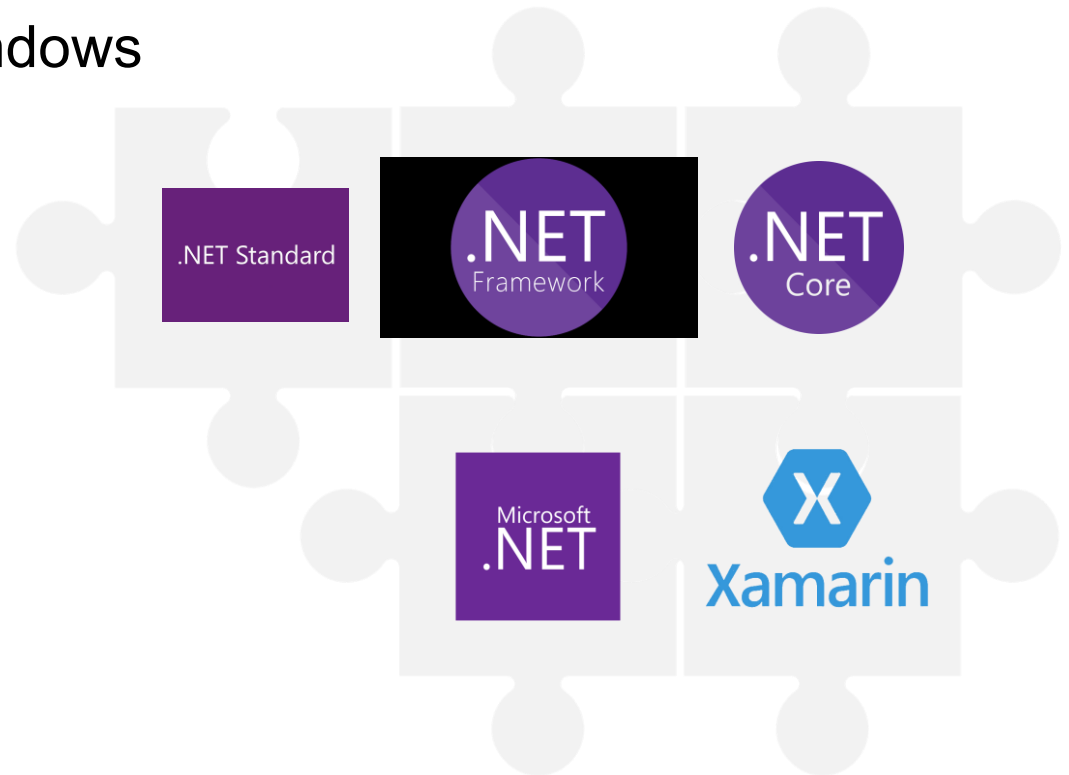
# What is .NET?

- .NET is a free, cross-platform, open-source developer platform for building many kinds of applications.
- It can run programs written in multiple languages, with C# being the most popular.



# Platforms

- .NET – runs everywhere
- .NET Framework – Windows
- Xamarin/Mono - mobile



# .NET Differences



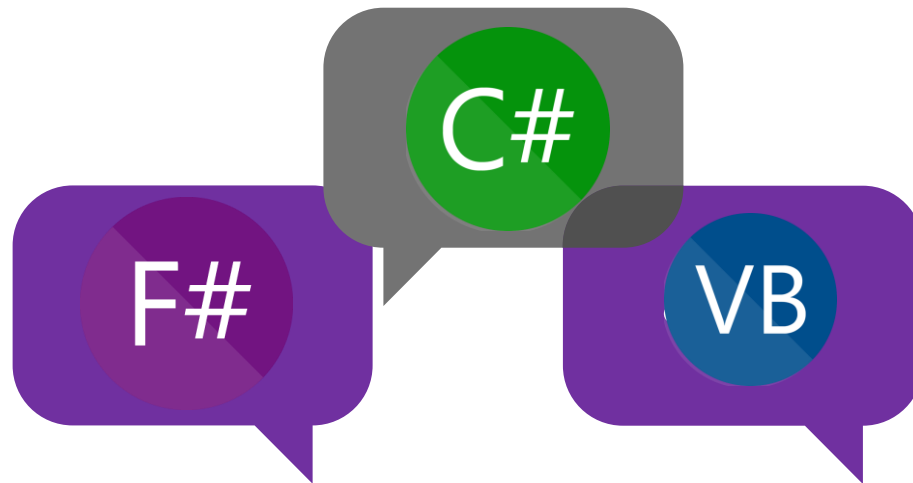
|                                                                                                  |                                                         |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Runs on Linux, macOS, & Windows                                                                  | Only runs on Windows                                    |
| Open-source & accepts contributions from community                                               | Source code available but does not accept contributions |
| All the innovation happens here! Supports more application types and delivers higher performance | Security & reliability bug fixes only                   |
| Not shipped with operating system                                                                | Included in Windows & updated by Windows updates        |

✓ Recommended



# Languages

- Microsoft offers 3 languages on the .NET platform
  - C#, F# and Visual Basic.



# Types of applications in .NET

- Your choice!

## Build it with .NET



### Web

Build web apps and services for macOS, Windows, Linux, and Docker.



### Mobile and desktop

Use a single codebase to build native apps for Windows, macOS, iOS, and Android.



### Cloud

Build scalable and resilient cloud-native apps that run on all major cloud providers.



### Artificial Intelligence and ML

Build smart apps with C#, OpenAI, and Azure.

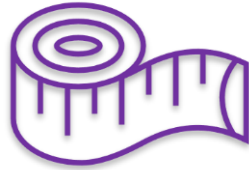
Microservices → Game development → Internet of Things → Mobile → Desktop → Front-end web → Back-end APIs → Data →



# .NET – strong points



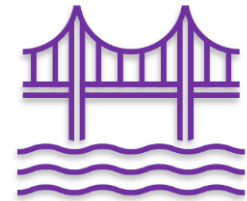
**Performance**



**Scalability**



**Security**



**Cross platform**



**Easy updates**



**Cloud friendly**



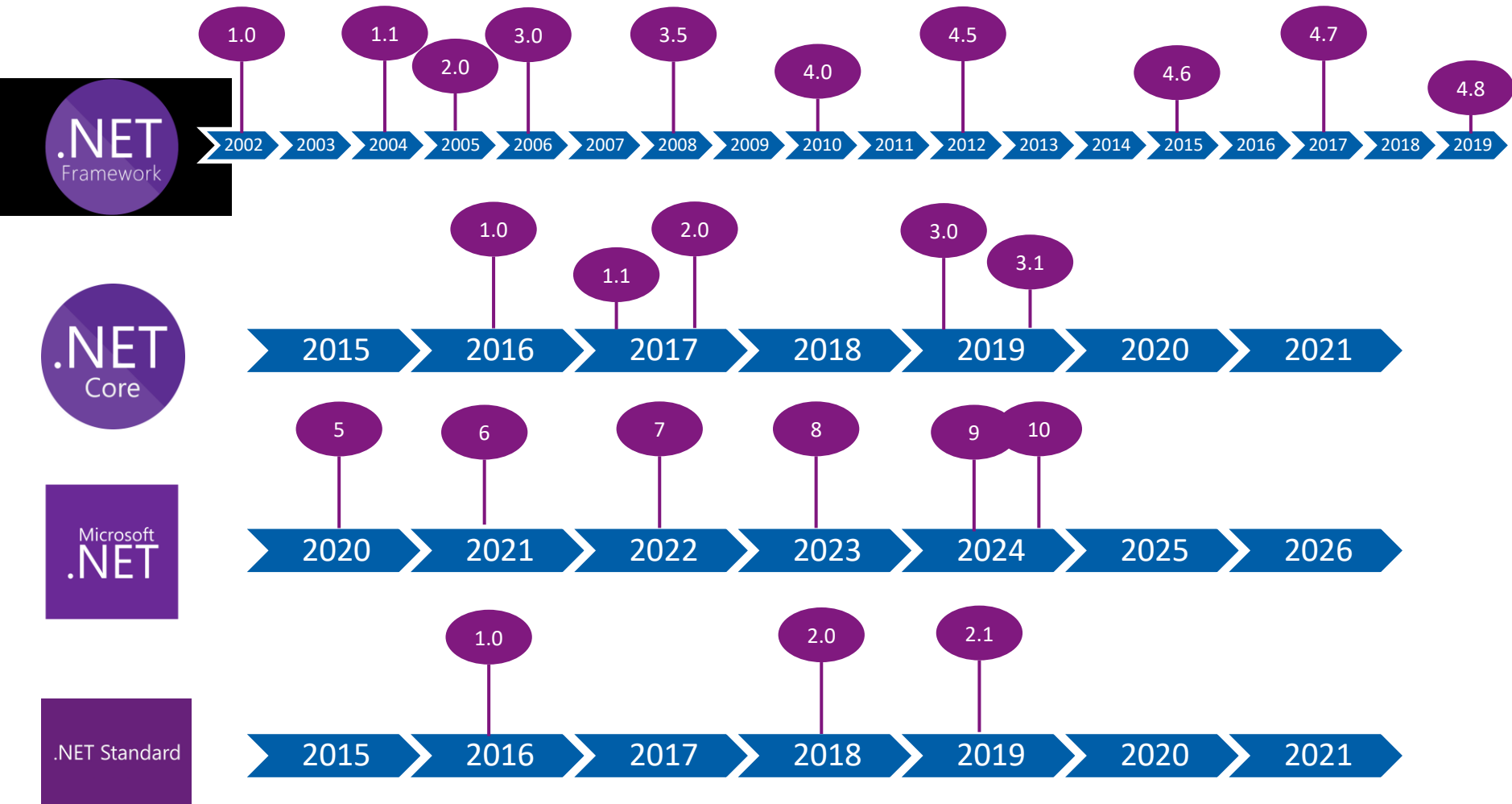
**Fast and open  
source**



**Build in  
Dependency  
Injection**



# The evolution of .NET



# How do we start using .NET?

1. Choose Your Development Environment
2. Install the .NET SDK
3. Select an IDE
4. Create Your First Project
5. Learn the Basics
6. Explore Libraries and Frameworks
7. Version Control

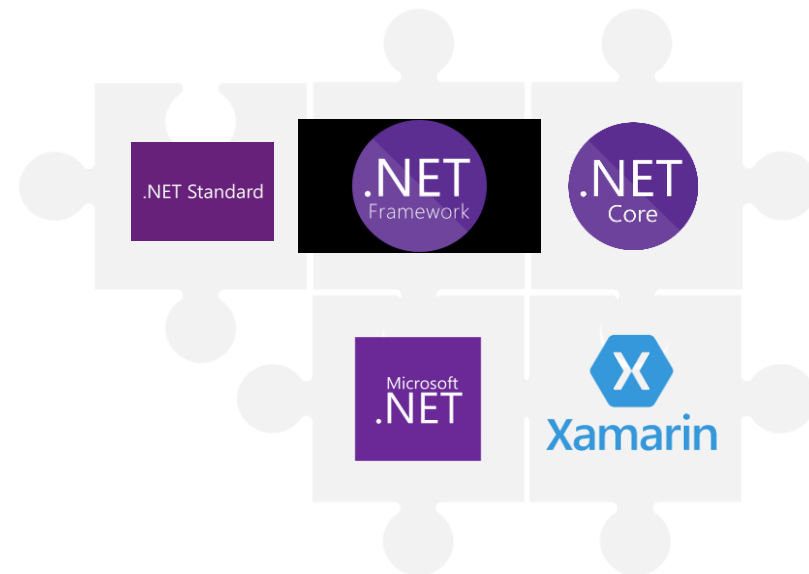


# .NET SDK

- Downloads:

<https://dotnet.microsoft.com/en-us/download/visual-studio-sdks>

- SDKs vs Runtime

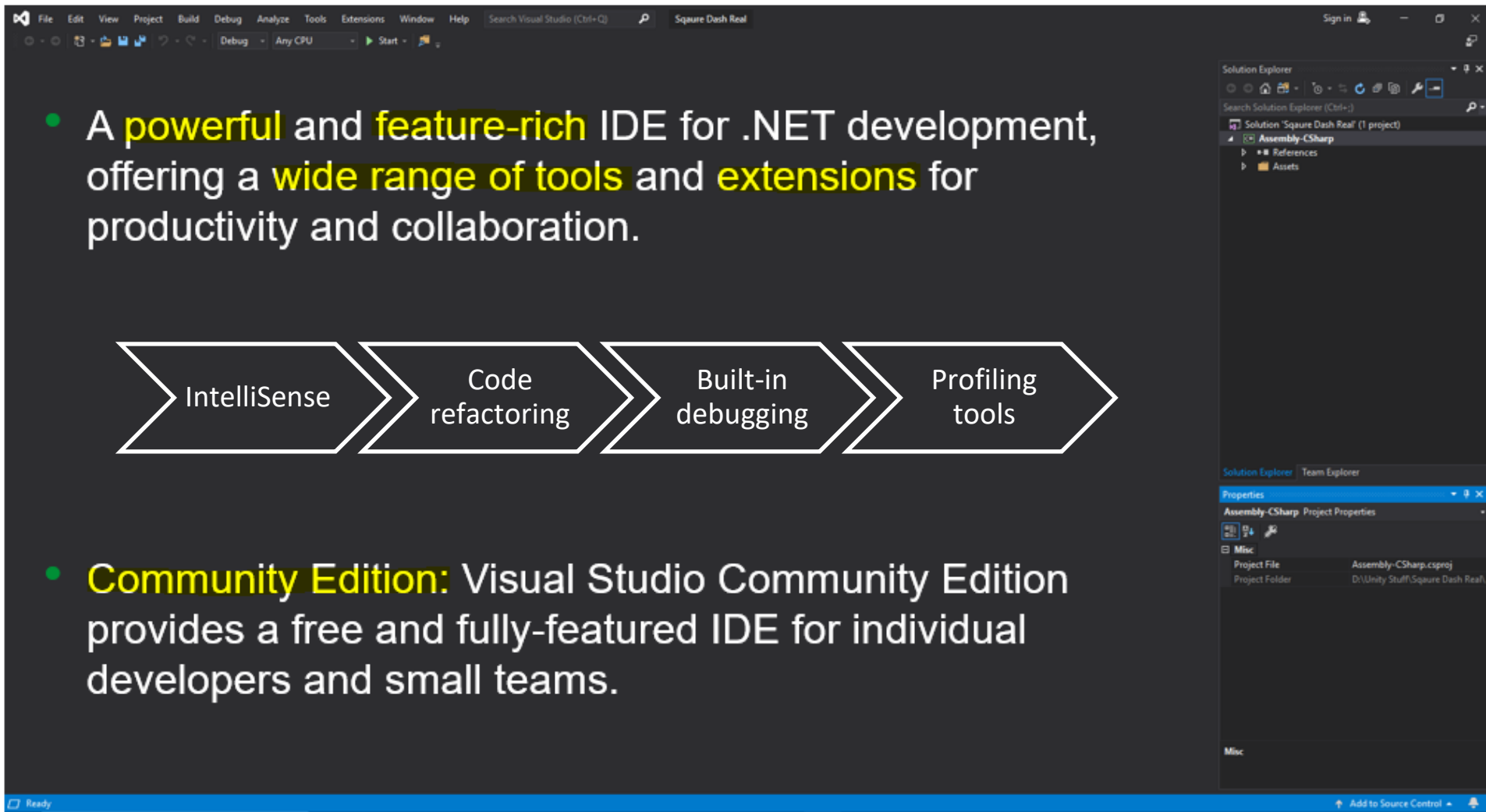


# IDE/Visual Studio

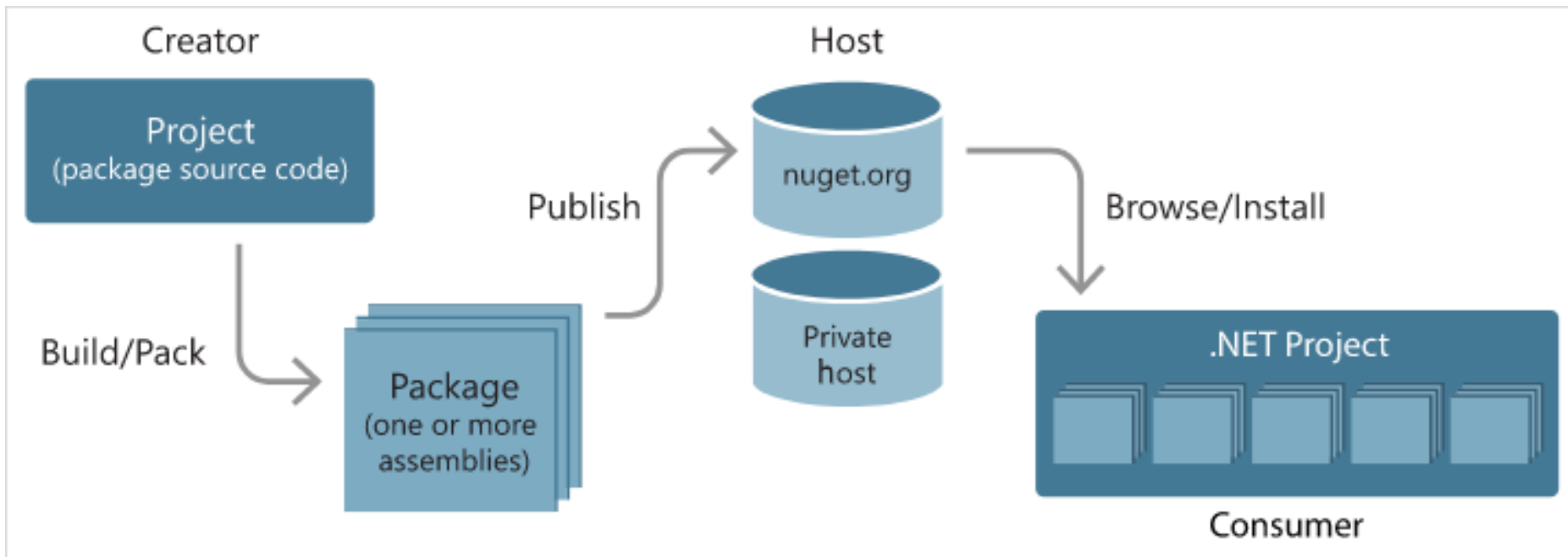
- A **powerful** and **feature-rich** IDE for .NET development, offering a **wide range of tools** and **extensions** for productivity and collaboration.



- **Community Edition:** Visual Studio Community Edition provides a free and fully-featured IDE for individual developers and small teams.

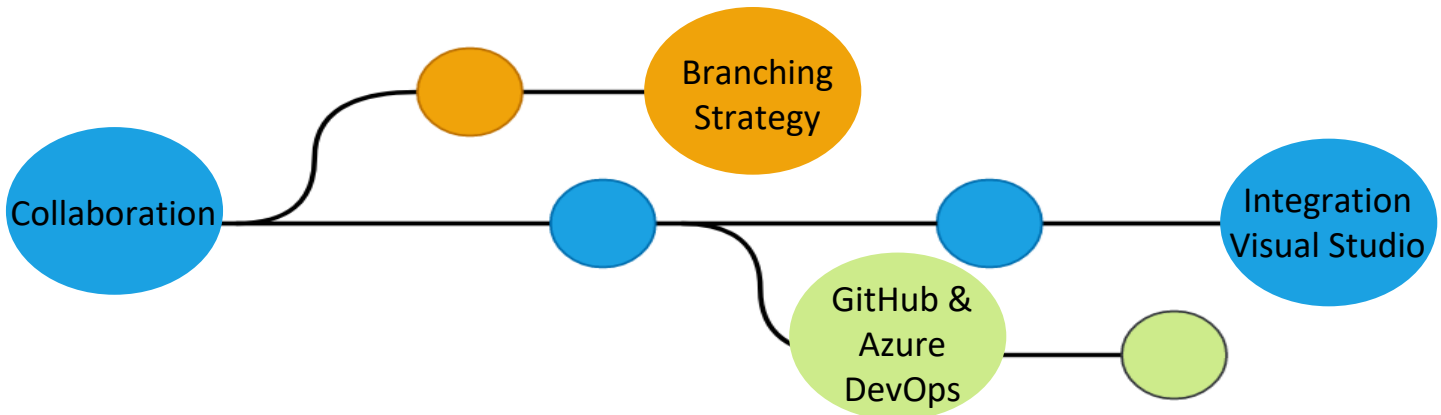


# NuGet Packages



# Git

- Git is a distributed version control system widely used in software development, including .NET projects.



- Best Practices: Adopting Git best practices such as feature branching, pull request reviews, and automated testing can enhance the stability and quality of .NET applications.



# How do we learn more about .NET?

- .NET website: <https://aka.ms/dotnet/website>
- .NET Documentation: <https://aka.ms/dotnet/documentation>

